

MUHAMMAD SHAHZAD IQBAL, PhD

Assistant Professor Candidate | NGS Expert | Genomics & Bioinformatics

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Portfolio: <https://shzadiqbal.github.io/mshahzadiqbal.github.io/>

Google Scholar: 402 citations | h-index: 11

PROFESSIONAL SUMMARY

Molecular biologist and bioinformatician with deep hands-on expertise in **Next-Generation Sequencing (NGS)** and large-scale transcriptomic analysis. My PhD research involved comprehensive RNA-Seq analysis of a complex polyploid crop, including development of a novel computational pipeline to correct polyploidy effects — work that is now in final publishable form and scheduled for submission to *Plant Molecular Biology* (Springer Nature). I have independently analyzed 50+ NGS projects (WGS, WES, RNA-seq, smRNA-seq), trained over 50 researchers, and supervised 28 M.Phil. students. Seeking an Assistant Professor position to advance NGS-driven genomics research and teaching.

RESEARCH INTERESTS

- Next-Generation Sequencing (NGS) data analysis & reproducible pipeline development
- Transcriptomics and differential gene expression in complex genomes
- Host–pathogen interactions and molecular virology
- Computational prediction of host-derived microRNAs against viruses
- Multi-omics data integration and functional genomics

EDUCATION

PhD in Molecular Biology (Major NGS Project)

Centre of Excellence in Molecular Biology (CEMB), University of the Punjab, Pakistan | 2021

Thesis: Transcriptomic expression behaviour of sugarcane cultivars against red rot disease.

Conducted high-throughput RNA-Seq analysis of multiple sugarcane cultivars to identify pathogen-responsive genes and defense mechanisms in a highly polyploid genome. Developed a **novel bioinformatics pipeline** specifically designed to minimize the confounding effects of polyploidy on differential expression analysis. This work represents substantial original contribution to NGS methodology for complex plant genomes.

Current Status: Manuscript submitted to *Plant Molecular Biology* (Springer Nature).

M.Phil. in Molecular Biology

CEMB, University of the Punjab, Pakistan | 2013

Thesis: In Vitro knockdown of PVY through shRNA (RNA interference & molecular virology).

BS (Hons) Biochemistry

The University of Lahore, Pakistan | 2011

PROFESSIONAL EXPERIENCE

Lecturer (Biochemistry)

University of Okara, Pakistan | August 2022 – Present

- Deputy Director, Office of Research, Innovation & Commercialization (ORIC)
- In-Charge, Department of Food Sciences and Technology
- Supervising 2 PhD + 28 completed M.Phil. + 40+ BS students
- Teaching Biochemistry, Molecular Biology, Bioinformatics and NGS technologies
- Trained 50+ researchers in NGS data analysis and computational biology

Assistant Professor (Biotechnology)

University of Central Punjab | Sep 2021 – Aug 2022

Research supervision and teaching in genetic engineering and molecular diagnostics.

Senior Lecturer (Molecular Biology & Bioinformatics)

The University of Lahore | Mar 2020 – Sep 2021

Managed computational server for genomics research. Taught bioinformatics and NGS.

NGS Application Scientist / Regional Manager

Molecular Products Co. (BGI Pakistan) | May 2018 – Feb 2020

- Independently analyzed **50+ NGS projects** (WGS, WES, RNA-seq, smRNA-seq, lncRNA-seq)
- Developed and optimized reproducible pipelines for variant calling and differential expression
- Delivered technical training and project design support to university research groups

University Research Associate (IRSIP Fellow)

QAAFI, The University of Queensland, Australia | Jul 2017 – Jan 2018

Supervisor: Prof. Robert Henry. Performed large-scale sugarcane RNA-Seq analysis (72+ samples) focused on red rot responsive genes — directly complementary to PhD research.

SELECTED PUBLICATIONS

Total Citations: 402 | h-index: 11 | Full list available on Google Scholar

- **Iqbal MS**, Sharif MN, Ali T, Henry RJ, Nasir IA. Comparative Transcriptomics identifies red rot induced defense response in sugarcane. *Manuscript ready for submission to Plant Molecular Biology (Springer Nature)*.
- Sharif MN, **Iqbal MS**, et al. (2022). Silencing of multiple target genes via ingestion of dsRNA... in *Helicoverpa armigera*. *Scientific Reports*. Citations: 43
- **Iqbal MS**, et al. (2017). In Silico MCMV Silencing Concludes Potential Host-Derived miRNAs in Maize. *Frontiers in Plant Science*. Citations: 41
- Hassan M, **Iqbal MS**, et al. (2022). Prediction of Site Directed miRNAs... Against Influenza C Virus. *Frontiers in Molecular Biosciences*.
- Ashraf MA, et al. (including **Iqbal MS**) (2022). In silico identification of sugarcane genome encoded microRNAs targeting sugarcane bacilliform virus. *PLoS ONE*.
- Jabbar B, **Iqbal MS**, et al. (2019). Target prediction of candidate miRNAs from *Oryza sativa* for silencing the RYMV genome. *Computational Biology and Chemistry*.

RESEARCH SUPERVISION

PhD: 2 (Ongoing) | **M.Phil.:** 28 (Completed) | **BS:** 40+

Majority of postgraduate projects focused on computational viral bioinformatics and host-derived microRNA prediction.

TECHNICAL SKILLS

NGS & Bioinformatics: RNA-Seq (EdgeR, DESeq2), WGS, WES, Variant Calling, Differential Expression, Virus detection from RNA-seq, GO/KEGG Pathway Analysis, De novo Assembly, Custom pipeline development for polyploid genomes

Programming & Computing: R (Advanced), Python, Linux, High-Performance Computing (HPC), Reproducible workflows

Molecular Biology: qRT-PCR, advanced PCR, DNA/RNA extraction, Gene cloning & transformation, Sanger sequencing

AWARDS & INTERNATIONAL EXPOSURE

- IRSIP Training Fellowship – 6 months genomics research at The University of Queensland, Australia
- British Council & HEC Study Visit to Scotland (Edinburgh, Glasgow, Dundee, Heriot-Watt)
- HEC Indigenous PhD Scholarship (4-year full funding)

References available upon request • Portfolio: <https://shzadiqbal.github.io/mshahzadiqbal.github.io/> • Google Scholar: <https://scholar.google.com/citations?user=-gzVrr4AAAAJ>